



Powering the UTWind VAWT

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Project Background

Commercial generators are optimized for high speeds, which limits **Vertical Axis Wind Turbine (VAWT)** performance through **high starting torque** and **reduced efficiency at 100-200 RPM**. This work develops a **custom generator** for the **UTWind 2026 turbine**, optimized for **low starting torque** and **improved efficiency at low RPM**. It establishes a **proof-of-concept** for future in-house generator development.

Key Term:

Starting Torque: Minimum torque to overcome startup resistance

Design Methodology

1

Brainstorming

- Four architectures were evaluated.

2

Decision Matrix

- Induction & SRG eliminated - poor low-RPM efficiency and high torque ripple.
- AFPMG eliminated - complex manufacturing and poor cogging mitigation.

3

Optimization Tables - Morph Chart

- Swept slot/pole, arc ratio, air gap, and pole segmentation

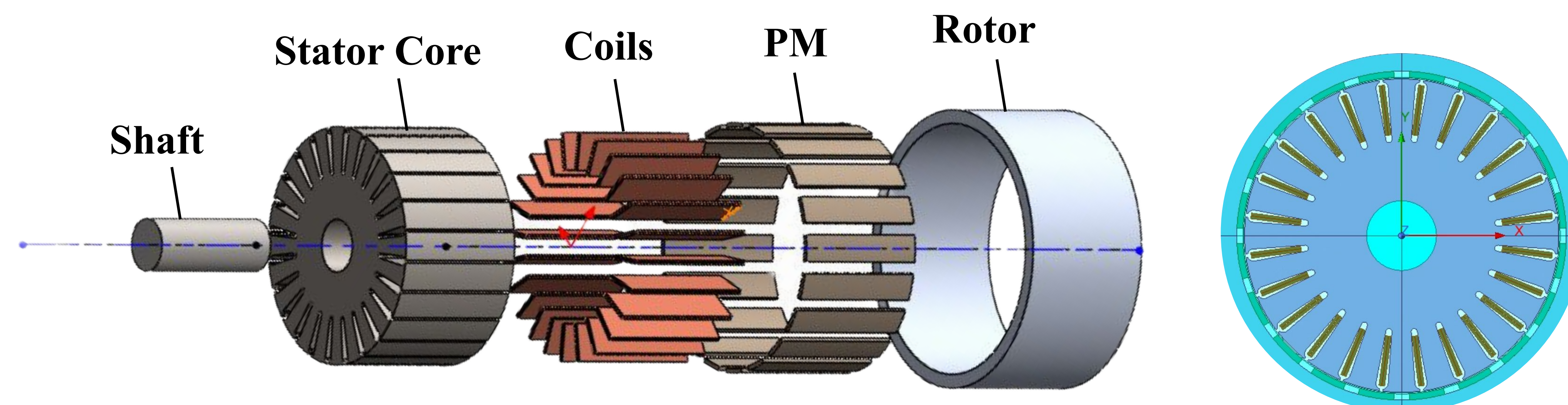
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FEA Validation

- Developed an initial analytical model using ANSYS RMxprt
- Validated electromagnetic performance in Maxwell 2D

Design Overview

Direct-drive Outer-Rotor Radial-Flux Permanent Magnet Synchronous Generator (RF-PMSG)



Key Geometry

- Slot / Pole:** 24/20
- Axial Length:** 100mm
- Air Gap:** ~1.0mm baseline
- Rotor OD:** ~210 mm
- Winding:** 3-Phase fractional slot concentrated winding (FSCW)
- Magnets:** Segmented surface mounted NdFe35

Design Goals

0.1 Nm
Starting
Torque

600-1200W
Power Output

100-200 RPM
Operating
Speed

>90%
Efficiency

Performance

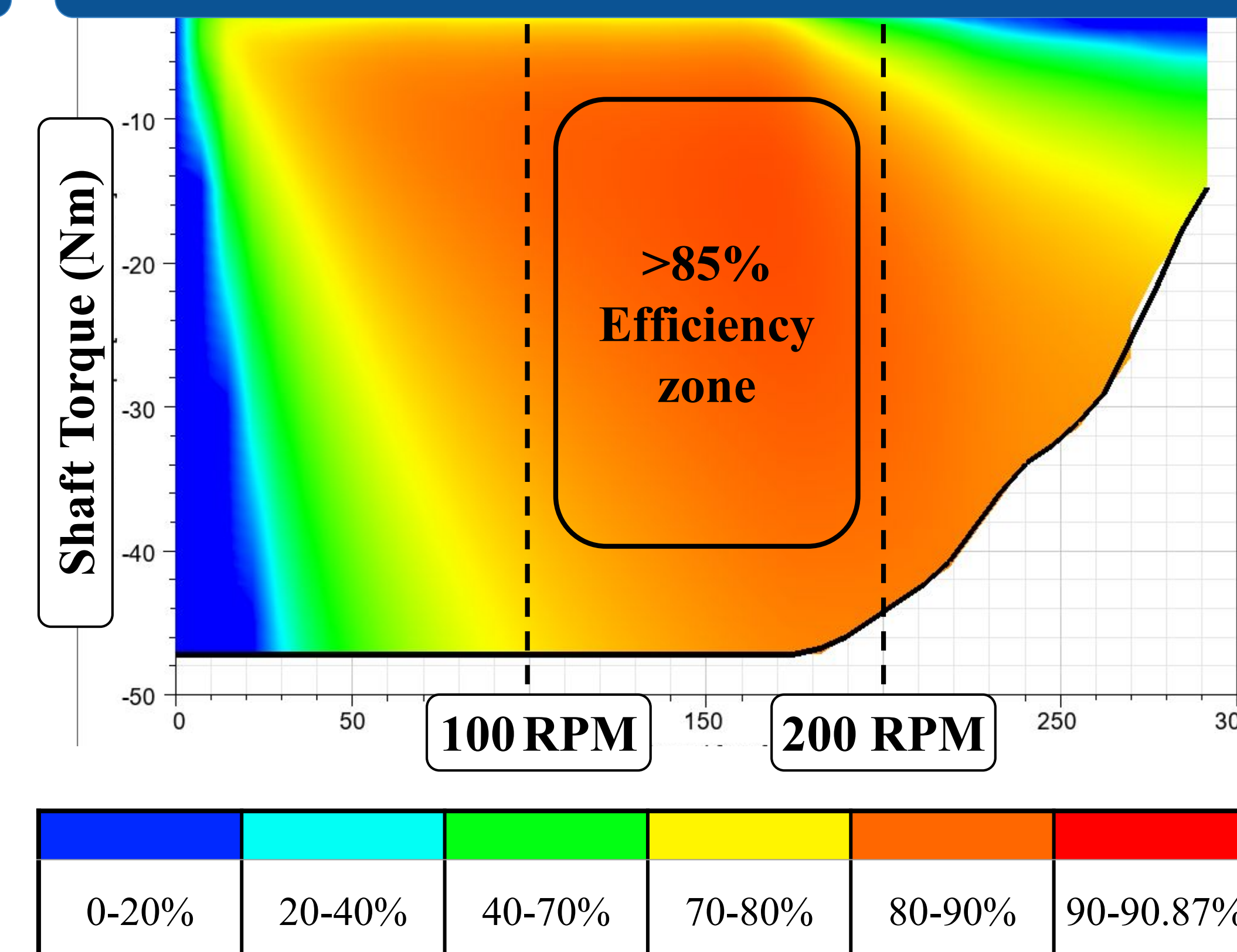
0.09 Nm
Peak Starting
Torque

842.8 W
Net Output
Power

100-200 RPM
Operating
Speed

67.06%
Efficiency

Efficiency Map



Next Steps

1

Phase 1 - Procurement

Source NdFe35 magnets & ASTM A677 (M19) lamination steel for rotor/stator fabrication

2

Phase 2 - Validation

Guided rotor-stator mating to maintain 1.0mm air gap; verify 600–1200W/0.1 N·m